

Retraction

Retracted: Acetochlor Affects Bighead Carp (*Aristichthys Nobilis*) by Producing Oxidative Stress, Lowering Tissue Proteins, and Inducing Genotoxicity

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This article has been retracted by Hindawi following an investigation undertaken by the publisher [1]. This investigation has uncovered evidence of one or more of the following indicators of systematic manipulation of the publication process:

- (1) Discrepancies in scope
- (2) Discrepancies in the description of the research reported
- (3) Discrepancies between the availability of data and the research described
- (4) Inappropriate citations
- (5) Incoherent, meaningless and/or irrelevant content included in the article
- (6) Manipulated or compromised peer review

The presence of these indicators undermines our confidence in the integrity of the article's content and we cannot, therefore, vouch for its reliability. Please note that this notice is intended solely to alert readers that the content of this article is unreliable. We have not investigated whether authors were aware of or involved in the systematic manipulation of the publication process.

Wiley and Hindawi regrets that the usual quality checks did not identify these issues before publication and have since put additional measures in place to safeguard research integrity.

We wish to credit our own Research Integrity and Research Publishing teams and anonymous and named external researchers and research integrity experts for contributing to this investigation.

The corresponding author, as the representative of all authors, has been given the opportunity to register their agreement or disagreement to this retraction. We have kept a record of any response received.

References

- [1] Y. Mahmood, R. Hussain, A. Ghaffar et al., "Acetochlor Affects Bighead Carp (*Aristichthys Nobilis*) by Producing Oxidative Stress, Lowering Tissue Proteins, and Inducing Genotoxicity," *BioMed Research International*, vol. 2022, Article ID 9140060, 12 pages, 2022.

Research Article

Acetochlor Affects Bighead Carp (*Aristichthys Nobilis*) by Producing Oxidative Stress, Lowering Tissue Proteins, and Inducing Genotoxicity

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Acetochlor is persistently used in the agroproduction sector to control broadleaf weeds. Due to frequent and continuous applications, this herbicide can reach nearby water bodies and may induce deleterious changes in aquatic life. Therefore, investigation of harmful impacts of different environmental pollutants, including herbicides, is vital to knowing the mechanisms of toxicity and devising control strategies. The current experiment included bighead carp ($n = 80$) to estimate adverse impacts. Fish were randomly placed in 4 different experimental groups (T0-T3) and were treated for 36 days with acetochlor at 0, 300, 400, and 500 $\mu\text{g/L}$. Fresh blood without any anticoagulant was obtained and processed for nuclear and morphological changes in erythrocytes. At the same time, various visceral organs, including the gills, liver, brain, and kidneys, were removed and processed on days 12, 24, and 36 to determine oxidative stress and various antioxidant biomarkers. Comet assays revealed significantly increased DNA damage in isolated cells of the liver, kidneys, brain, and gills of treated fish. We recorded increased morphological and nuclear changes ($P \leq 0.05$) in the erythrocyte of treated fish. The results on oxidative stress showed a higher quantity of oxidative biomarkers and a significantly ($P \leq 0.05$) low concentration of cellular proteins in the gills, liver, brain, and kidneys of treated fish compared to unexposed fish. Our research findings concluded that acetochlor renders oxidative stress in bighead carp.

1. Introduction

Insecticides/pesticides especially herbicides have become a severe threat in the last few years. These have become a serious hazard to health because of their uncontrolled use in the aquatic environment and agriculture [1–6]. In the marine and terrestrial ecosystem, various ailments are caused in dif-

ferent animals due to unintended exposure to insecticides/pesticides, herbicides, and fungicides [7–11]. Various earlier reports have highlighted that aquatic species are mainly and extensively susceptible to several natural and synthetic toxicants than terrestrial animals because of the entry of insecticides/pesticides from agriculture, production sites/industries into water [12–15]. The use of pesticides in agriculture



FIGURE 1: Various organs of the fish treated with acetochlor showing normal brain, congested gills, and kidneys.

results in diffusing pollution where many soils, contaminating materials, leach to groundwater and ultimately reach drinking water. Since using several pesticides is a common practice, their leaching leads to the risk up to an alarming level [5].

Several studies have recorded that disclosure to synthetic composites, including herbicides, and by-products of various antiseptics, pesticides, fluorinated substances, and insecticides [7, 9, 16] cause deleterious effects to the aquatic life. It is also observed that even exposure to residues of natural and synthetic chemical compounds via the food chain renders abnormalities in multiple tissues of animals/fish, ultimately leading to disruption of several metabolic processes [4, 11, 17–20]. Aquatic animals, particularly fish, are considered the most susceptible to insecticides/pesticides and are reliable test specimens for the evaluation of the quality of the aquatic environment [21–24].

Bighead carp (*Aristichthys nobilis*) is commonly found in Pakistan's rivers and freshwater lakes and is a cultivable fish species. There are different scientific synonyms of bighead carp, such as *Cephalus hypothalamus* (Hong Kong), *Leuciscus nobilis* (Canton), *Hypophthalmichthys mandschuricus* (Shanghai), and *Hypophthalmichthys simony*. *Hypophthalmichthys nobilis* has another name like *Aristichthys nobilis* based on the divergent form of a branchial row, pharyngeal dentition, and length of the abdominal keel. It is demonstrated that various environmental contaminants like herbicides and insecticides mainly induce toxic effects via rapid induction of oxidative stress leading to depletion of antioxidant biomarkers in exposed animals [25–29]. The evaluation of hematological and biochemical biomarkers acts as useful and reliable bioindicators of toxicity in aquatic animals [3, 9, 30]. Moreover, serum biochemistry and different biomarkers in visceral tissues, such as various enzymes and proteins, are routinely used to assess health of aquatic animals [11, 31, 32].

The toxicity of acetochlor on behavior, growth and reproduction, and oxidative stress in different organisms have been reported [24, 33]. Various studies have investigated that numerous synthetic and natural pollutants disrupt the normal physiological processes of cells by interfering with various cellular proteins (p38 mitogen-activated protein, C-reactive protein, and G protein-coupled receptors), hormonal-signaling pathways, redox-sensitive signal transduction pathways, kinase and transcription factor AP-1 leading to induction of inflammation, and alterations in the blood and necrosis of various organs in exposed organisms [34, 35]. Moreover, it is also determined that different chemicals act as endocrine-disrupting compounds and cause abnormalities in redox homeostasis mechanisms, lowering cellular proteins/antioxidant enzymes, thus, mitochondrial dysfunctions and apoptosis [36, 37]. However, insufficient information is available about producing oxidative stress, lowering tissue proteins, and inducing genotoxicity due to chlorinated herbicides such as “acetochlor” in freshwater fish, especially bighead carp.

A chlorinated herbicide like acetochlor is commonly used on soybeans, maize, sugar beets, and different cereal crops to remove broadleaf weeds [38–40]. Acetochlor can enter the body of different aquatic animals via direct contact with contaminated water, ingestion, and dermal interaction resulting in physiological changes. Acetochlor is routinely used to control weeds in China and many other countries worldwide. It inhibits the growth of weeds at the early developmental stage by affecting cell integrity [41]. In China, acetochlor has been used for many years [24]. In 2010 and 2011, a study was carried out in North Carolina and Iowa states involving 33,484 people; however, acetochlor was used on 4,026 people, and there was a high probability of cancer among exposed people. Since the registration of acetochlor in 1994, it has become an herbicide of choice in the USA [42]. The organisms present in aquatic ecosystems are very

TABLE 1: Oxidative stress parameters and levels of cellular proteins/antioxidant enzymes present in liver tissue of bighead carp exposed to different levels of acetochlor.

Biochemical parameters/days	Groups/treatments			
	T0 (0.0 µg/L)	T1 (300 µg/L)	T2 (400 µg/L)	T3 (500 µg/L)
ROS (optical density)				
12	0.35 ± 0.03	0.36 ± 0.02	0.39 ± 0.03	0.83 ± 0.07*
24	0.36 ± 0.01	0.37 ± 0.01	0.66 ± 0.02*	0.85 ± 0.05*
36	0.38 ± 0.04	0.41 ± 0.03	0.73 ± 0.02*	0.91 ± 0.05*
TBARS (nmol/TBARS formed/mg protein/min)				
12	37.41 ± 2.93	39.22 ± 1.95	41.06 ± 1.15	54.92 ± 3.74*
24	38.88 ± 1.94	40.67 ± 2.95	50.43 ± 1.97*	56.24 ± 2.92*
36	40.01 ± 3.94	41.94 ± 2.93	51.87 ± 2.34*	58.82 ± 4.53*
Reduced GSH (µmol/g tissue)				
12	8.65 ± 1.17	7.64 ± 0.06	6.63 ± 0.13	5.62 ± 0.15*
24	8.37 ± 1.12	7.43 ± 1.10	6.04 ± 0.02*	5.55 ± 0.12*
36	8.33 ± 1.14	7.37 ± 1.05	5.92 ± 1.15*	5.47 ± 0.14*
Cellular proteins/antioxidant enzymes				
SOD (units/mg protein)				
12	11.64 ± 0.12	10.96 ± 0.12	10.28 ± 0.13	10.26 ± 0.11
24	11.65 ± 0.15	10.31 ± 0.15	9.45 ± 0.15*	7.57 ± 0.14*
36	10.74 ± 0.18	9.67 ± 0.18	7.09 ± 0.18*	7.03 ± 0.17*
CAT (units/min)				
12	8.69 ± 0.19	7.51 ± 0.19	7.13 ± 0.19	5.15 ± 0.19*
24	7.98 ± 0.17	7.34 ± 0.16	5.62 ± 0.16*	5.02 ± 0.16*
36	7.94 ± 0.16	7.01 ± 0.16	5.45 ± 0.16*	4.94 ± 0.16*
POD (units/µg)				
12	4.07 ± 0.09	3.53 ± 0.09	2.77 ± 0.09*	2.46 ± 0.09*
24	4.02 ± 0.08	3.46 ± 0.09	2.62 ± 0.09*	2.40 ± 0.09*
36	3.95 ± 0.09	3.38 ± 0.09	2.54 ± 0.09*	2.39 ± 0.09*

Mean ± SE values with asterisks in a row vary significantly ($P \leq 0.05$) as compared to the control group.

delicate and diversified, and almost all reflect the exact physical and biochemical changes when exposed to various toxicants [24, 26, 43]. As discussed above, herbicides render various biochemical alternations, especially oxidative stress in aquatic species, and no comprehensive study was available about bighead carp; hence, we planned this research to understand the development of oxidative stress caused by acetochlor in general, but in particular how oxidative stress behaves in body tissues of the *Aristichthys nobilis* (bighead carp).

2. Materials and Methods

2.1. Chemicals. Acetochlor was procured from the local commercial market (M/S Ali Akbar Enterprises, Pakistan) district Lodhran, Pakistan. Different other experimental chemicals of analytical grade were purchased from Sigma Aldrich (USA) and Merck (Germany). To estimate serum biochemical parameters, we obtained different commercial kits from Randox Company (Pvt.) Pakistan.

2.2. Experimental Species and Management. This study was conducted on bighead carp ($n = 80$) freshwater fish (*Aristichthys nobilis*) obtained from a local commercial fish farm. All the fish had uniform size, body weight (140 - 155 g), and age. The experimental test specimens were transferred to the laboratory in plastic bags with adequate oxygen. We kept all experimental specimens in aquaria made of glass (14" L, 10" W, and 12" H) for 10 days to allow them to acclimatize. All experimental fish were fed commercial feed at 2-3% of body weight twice daily, i.e., morning and evening. All aquaria had their residual feed and fecal contents eliminated.

2.3. Experimental Groups. After acclimatizing the fish, all experimental fish ($n = 80$) were randomly divided into four equal groups (T0-T3). Each aquarium was having 100 L water carrying capacity. Group T0 served as the control group, whereas fish of groups T1, T2, and T3 were treated with acetochlor at 300, 400, and 500 µg/L for 36 days, respectively. During the entire experiment, residual feed and fecal elements were drained and removed daily. All experimental test specimens

TABLE 2: Oxidative stress parameters and levels of cellular proteins/antioxidant enzymes in kidney tissues of bighead carp exposed to different levels of acetochlor.

Biochemical parameters/days	Groups/treatments			
	T0 (0.0 µg/L)	T1 (300 µg/L)	T2 (400 µg/L)	T3 (500 µg/L)
ROS (optical density)				
12	0.49 ± 0.01	0.56 ± 0.01	0.62 ± 0.01*	0.70 ± 0.01*
24	0.52 ± 0.01	0.58 ± 0.01	0.64 ± 0.01*	0.73 ± 0.01*
36	0.55 ± 0.01	0.61 ± 0.01	0.69 ± 0.01*	0.77 ± 0.01*
TBARS (nmol/TBARS formed/mg protein/min)				
12	28.50 ± 0.5	32.20 ± 0.5	35.91 ± 0.5*	39.62 ± 0.5*
24	29.07 ± 0.6	33.07 ± 0.6	37.06 ± 0.6*	41.05 ± 0.6*
36	29.71 ± 0.7	33.78 ± 0.7	37.78 ± 0.7*	41.82 ± 0.7*
Reduced GSH (µmol/g tissue)				
12	7.69 ± 0.3	6.44 ± 0.3	5.20 ± 0.2*	3.94 ± 0.2*
24	7.61 ± 0.3	6.39 ± 0.2	5.17 ± 0.2*	3.88 ± 0.2*
36	7.55 ± 0.3	6.34 ± 0.2	5.09 ± 0.2*	3.84 ± 0.2*
Cellular proteins/antioxidant enzymes				
SOD (units/mg protein)				
12	15.50 ± 0.32	13.52 ± 0.3	11.55 ± 0.32*	9.54 ± 0.32*
24	15.39 ± 0.35	13.13 ± 0.3	10.98 ± 0.35*	8.76 ± 0.35*
36	15.25 ± 0.36	13.09 ± 0.3	10.85 ± 0.36*	8.67 ± 0.36*
CAT (units/min)				
12	5.11 ± 0.1	4.57 ± 0.09	4.03 ± 0.09	3.45 ± 0.07*
24	5.07 ± 0.1	4.55 ± 0.09	3.97 ± 0.08*	3.42 ± 0.07*
36	5.03 ± 0.1	4.44 ± 0.09	3.91 ± 0.08*	3.29 ± 0.06*
POD (units/µg)				
12	5.97 ± 0.13	5.32 ± 0.13	4.66 ± 0.11	4.03 ± 0.10*
24	5.91 ± 0.13	5.29 ± 0.12	4.62 ± 0.11*	3.94 ± 0.10*
36	5.88 ± 0.13	5.26 ± 0.12	4.59 ± 0.10*	3.88 ± 0.10*

Mean ± SE values with asterisks in a row vary significantly ($P \leq 0.05$) as compared to the control group.

were carefully observed twice daily for any noticeable clinical and behavioral ailments.

2.4. Blood Analysis and Genotoxicity Assessment. Blood (2.5 mL) was drawn from the caudal vein of each fish with the help of a hypodermic needle (26 gauge) on days 12, 24, and 36 [8]. A thin blood smear was made from each fish's fresh blood without anticoagulant to examine nuclear and morphological changes in erythrocytes. Blood smears were dried right away, fixed with 100% methyl alcohol, and stained with Giemsa's stain. A light microscope with an oil immersion lens was used to examine 1500 erythrocytes from each fish, aided by a computer [8].

Under alkaline conditions, Comet assay or single-cell gel electrophoresis was used to estimate DNA damage in various tissues, i.e., the gills, brain, liver, and kidneys [1, 4]. After dissecting, these tissues were isolated and immersed separately in chilled normal saline solution, homogenized, and centrifuged (0.2g). Every tissue's supernatant with suspended single cells was isolated and subjected to single-cell gel electrophoresis or Comet assay

[1]. Briefly, agarose with normal point (1%) and low melting point (1%) was dissolved in Milli-Q water and prepared thin smears on frosted glass slides [1]. After preparation, the cells present on slides were lysed in a cold buffer solution. Then, the slides were electrophoresed in a horizontal tank with a refrigerated electrophoresis solution at 25 V for 30 minutes [9]. After electrophoresis, the slides were neutralized using an ice-cold 0.4 M tris buffer (pH 7.5). Finally, ethidium bromide-stained slides were examined under a fluorescence microscope at a magnification of 40x. A total of 500 cells/fish/slide were observed to estimate the occurrence of damaged DNA in each slide and tissue sample.

2.5. Tissues Biochemical Changes. Homogenates from tissues of gills, liver, brain, and kidneys were prepared and subjected to the determination of various biochemical parameters including ROS [44], TBARS [45], and GSH [46], similarly, from the homogenates assessed cellular proteins/antioxidant enzymes including POD [47], SOD [48], and CAT [47].

TABLE 3: Oxidative stress parameters and levels of tissue proteins/antioxidant enzyme in gill tissue of bighead carp treated with various levels of acetochlor.

Biochemical parameters/days	Groups/treatments			
	T0 (0.0 $\mu\text{g/L}$)	T1 (300 $\mu\text{g/L}$)	T2 (400 $\mu\text{g/L}$)	T3 (500 $\mu\text{g/L}$)
ROS (optical density)				
12	0.32 $\pm$ 0.01	0.38 $\pm$ 0.01	0.46 $\pm$ 0.02*	0.53 $\pm$ 0.04*
24	0.34 $\pm$ 0.01	0.41 $\pm$ 0.02	0.49 $\pm$ 0.02*	0.56 $\pm$ 0.04*
36	0.36 $\pm$ 0.01	0.43 $\pm$ 0.02	0.51 $\pm$ 0.03*	0.58 $\pm$ 0.04*
TBARS (nmol/TBARS formed/mg protein/min)				
12	40.66 $\pm$ 0.60	44.46 $\pm$ 0.62	48.26 $\pm$ 0.61*	52.06 $\pm$ 0.62*
24	41.12 $\pm$ 0.61	44.87 $\pm$ 0.61	48.63 $\pm$ 0.61*	52.38 $\pm$ 0.63*
36	41.19 $\pm$ 0.61	45.03 $\pm$ 0.62	48.81 $\pm$ 0.62*	52.69 $\pm$ 0.63*
Reduced GSH ($\mu\text{mol/g}$ tissue)				
12	2.57 $\pm$ 0.06	2.23 $\pm$ 0.06	1.89 $\pm$ 0.06*	1.55 $\pm$ 0.06*
24	2.43 $\pm$ 0.05	2.12 $\pm$ 0.05	1.82 $\pm$ 0.05*	1.51 $\pm$ 0.05*
36	2.33 $\pm$ 0.05	2.04 $\pm$ 0.05	1.75 $\pm$ 0.05*	1.46 $\pm$ 0.05*
Cellular proteins/antioxidant enzymes				
SOD (units/mg protein)				
12	10.78 $\pm$ 0.2	9.72 $\pm$ 0.2	8.67 $\pm$ 0.2	7.20 $\pm$ 0.1*
24	10.67 $\pm$ 0.2	9.52 $\pm$ 0.2	8.36 $\pm$ 0.2	7.21 $\pm$ 0.1*
36	10.56 $\pm$ 0.2	9.31 $\pm$ 0.2	8.06 $\pm$ 0.1	6.82 $\pm$ 0.1*
CAT (units/min)				
12	3.02 $\pm$ 0.04	2.74 $\pm$ 0.05	2.47 $\pm$ 0.05	2.18 $\pm$ 0.04*
24	2.97 $\pm$ 0.05	2.68 $\pm$ 0.05	2.42 $\pm$ 0.05*	2.14 $\pm$ 0.05*
36	2.94 $\pm$ 0.05	2.66 $\pm$ 0.05	2.38 $\pm$ 0.05*	2.07 $\pm$ 0.05*
POD (units/ μg)				
12	0.40 $\pm$ 0.02	0.36 $\pm$ 0.02	0.31 $\pm$ 0.01*	0.25 $\pm$ 0.01*
24	0.39 $\pm$ 0.02	0.34 $\pm$ 0.02	0.29 $\pm$ 0.01*	0.23 $\pm$ 0.01*
36	0.38 $\pm$ 0.02	0.33 $\pm$ 0.02	0.28 $\pm$ 0.01	0.21 $\pm$ 0.01*

Mean $\pm$ SE values with asterisks in a row vary significantly ($P \leq 0.05$) as compared to the control group.

2.6. Statistical Analysis. The trial's research findings are reported as mean $\pm$ SE. The data from our study (all experiments) were analyzed using ANOVA with SPSS statistics (version 20) software, and the group means were compared using a post hoc Tukey's test. We set a significance level at $P \leq 0.05$.

3. Results

3.1. Physical Findings. At necropsy, different macroscopic lesions, including hyperemic and congested gills, edematous and congested kidneys, mild to moderate congestion in brain, hyperemic muscles, and congested and moderately friable liver were examined in fish of group T3 at days 24 and 36 of the study (Figure 1).

3.2. Reactive Oxygen Species and Cellular Proteins/Antioxidant Enzymes in the Liver. The concentration of ROS and TBARS in the liver significantly ($P \leq 0.05$) increased in the fish of groups T2 (400 $\mu\text{g/L}$) and T3 (500 $\mu\text{g/L}$) treated with acetochlor after experimental days

12, 24, and 36. GSH, SOD, CAT, and POD concentration dropped significantly ($P \leq 0.05$) in the liver of fish of groups T2 and T3 treated with 400 $\mu\text{g/L}$ and 500 $\mu\text{g/L}$ acetochlor after days 12, 24, and 36 of the experiment, respectively (Table 1).

3.3. Reactive Oxygen Species and Cellular Proteins/Antioxidant Enzymes in Kidneys. The quantity of ROS and TBARS increased significantly ($P \leq 0.05$) in the kidneys of fish of groups T2 (400 $\mu\text{g/L}$) and T3 (500 $\mu\text{g/L}$) treated with acetochlor after days 12, 24, and 36 of the experiment. GSH contents dropped significantly ($P \leq 0.05$) in the fish of groups T2 and T3 treated with acetochlor after days 12, 24, and 36 in kidneys (Table 2). SOD and CAT in the kidneys decreased significantly in the fish of groups T2 and T3 after days 12, 24, and 36. POD dropped significantly ($P \leq 0.05$) in the kidneys of fish of groups T2 and T3 after days 12, 24, and 36 (Table 2).

3.4. Reactive Oxygen Species and Cellular Proteins/Antioxidant Enzymes in Gills. The quantity of ROS and

TABLE 4: Oxidative stress parameters and levels of cellular proteins/antioxidant enzymes in brain tissues of bighead carp exposed to different levels of acetochlor.

Biochemical parameters/days	Groups/treatments			
	T0 (0.0 $\mu\text{g/L}$)	T1 (300 $\mu\text{g/L}$)	T2 (400 $\mu\text{g/L}$)	T3 (500 $\mu\text{g/L}$)
ROS (optical density)				
12	0.40 $\pm$ 0.02	0.50 $\pm$ 0.02	0.61 $\pm$ 0.03*	0.70 $\pm$ 0.03*
24	0.45 $\pm$ 0.02	0.54 $\pm$ 0.02	0.63 $\pm$ 0.03*	0.71 $\pm$ 0.04*
36	0.52 $\pm$ 0.02	0.60 $\pm$ 0.03	0.65 $\pm$ 0.03*	0.75 $\pm$ 0.04*
TBARS (nmol/TBARS formed/mg protein/min)				
12	18.59 $\pm$ 0.4	22.28 $\pm$ 0.4	25.96 $\pm$ 0.5*	29.65 $\pm$ 0.6*
24	19.07 $\pm$ 0.4	22.76 $\pm$ 0.4	26.45 $\pm$ 0.5*	30.13 $\pm$ 0.6*
36	19.17 $\pm$ 0.4	22.91 $\pm$ 0.5	26.65 $\pm$ 0.6*	30.39 $\pm$ 0.6*
Reduced GSH ($\mu\text{mol/g}$ tissue)				
12	2.99 $\pm$ 0.06	2.59 $\pm$ 0.06	2.18 $\pm$ 0.05*	1.78 $\pm$ 0.04*
24	2.95 $\pm$ 0.06	2.55 $\pm$ 0.05	2.15 $\pm$ 0.04*	1.76 $\pm$ 0.03*
36	2.87 $\pm$ 0.06	2.49 $\pm$ 0.05	2.09 $\pm$ 0.04*	1.72 $\pm$ 0.03*
Cellular proteins/antioxidant enzymes				
SOD (units/mg protein)				
12	13.64 $\pm$ 0.3	12.08 $\pm$ 0.3	10.51 $\pm$ 0.2*	8.95 $\pm$ 0.2*
24	13.55 $\pm$ 0.3	11.99 $\pm$ 0.2	10.42 $\pm$ 0.2*	8.86 $\pm$ 0.1*
36	13.44 $\pm$ 0.3	11.87 $\pm$ 0.2	10.31 $\pm$ 0.2*	8.74 $\pm$ 0.1*
CAT (units/min)				
12	4.15 $\pm$ 0.08	3.65 $\pm$ 0.08	3.15 $\pm$ 0.07*	2.65 $\pm$ 0.06*
24	4.11 $\pm$ 0.08	3.57 $\pm$ 0.08	3.03 $\pm$ 0.07*	2.49 $\pm$ 0.06*
36	4.06 $\pm$ 0.08	3.51 $\pm$ 0.07	2.97 $\pm$ 0.07*	2.42 $\pm$ 0.06*
POD (units/ μg)				
12	3.13 $\pm$ 0.06	2.78 $\pm$ 0.06	2.43 $\pm$ 0.05*	2.08 $\pm$ 0.04*
24	3.10 $\pm$ 0.06	2.72 $\pm$ 0.05	2.34 $\pm$ 0.05*	1.97 $\pm$ 0.04*
36	3.03 $\pm$ 0.06	2.66 $\pm$ 0.05	2.30 $\pm$ 0.05*	1.93 $\pm$ 0.04*

Mean $\pm$ SE values with asterisks in a row vary significantly ($P \leq 0.05$) as compared to the control group.

TBARS increased significantly ($P \leq 0.05$) in the fish of groups T2 and T3 treated with 400 $\mu\text{g/L}$ 500 $\mu\text{g/L}$ acetochlor, respectively, after days 12, 24, and 36 of the experiment. GSH, SOD, CAT, and POD concentration dropped significantly ($P \leq 0.05$) in the fish of groups T2 and T3 treated with 400 $\mu\text{g/L}$ and 500 $\mu\text{g/L}$ acetochlor after days 12, 24, and 36 of the experiment, respectively (Table 3).

3.5. Reactive Oxygen Species and Cellular Proteins/Antioxidant Enzymes in the Brain. The quantity of ROS and TBARS in the brain increased significantly ($P \leq 0.05$) in the fish of groups T2 and T3 treated with 400 $\mu\text{g/L}$ acetochlor after days 12, 24, and 36 of the experiment. GSH contents, SOD, CAT, and POD decreased significantly ($P \leq 0.05$) in the brain of fish of groups T2 (400 $\mu\text{g/L}$) and T3 (500 $\mu\text{g/L}$) treated with acetochlor after days 12, 24, and 36 of the experiment (Table 4).

3.6. Genotoxicity. The results of nuclear and morphological changes in erythrocytes of treated fish showed significantly ($P \leq 0.05$) higher values of micronuclei, condensed nuclei,

lobed nuclei, red blood cells with a pear shape, and pear shape erythrocytes (Figure 2). A significantly ($P \leq 0.05$) increased frequency of DNA damage (Table 5) was recorded in isolated cells of the liver, kidneys, gills, and brain at different intervals of the experiment when compared with that of normal fish (Figure 3).

4. Discussion

In the present study, various oxidative stress parameters increased significantly in acetochlor-treated freshwater fish. Acetochlor is a toxic herbicide [49]. Increased oxidative stress could be caused by the toxic effect of acetochlor on different biological processes such as metabolism [50–52]. Many intrinsic and extrinsic factors are found to induce oxidative stress in different animals. Oxidative stress may be caused by an imbalance between ROS production and antioxidant defenses [51, 53]. Another possibility could be due to the reaction of toxicants with water to produce superoxide, which resultantly increased oxidative stress in freshwater fish [9, 50].

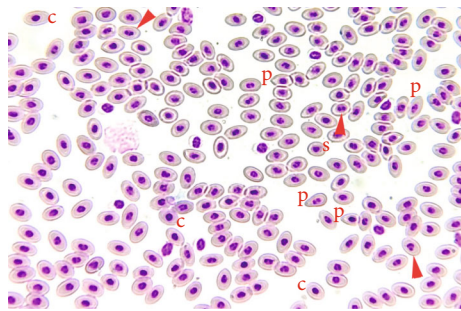


FIGURE 2: Blood smear of fish treated with acetochlor (500 µg/L) stained with Giemsa stain showing pear-shaped erythrocytes (p), spherocytes (s), condensed nuclei (c), and micronucleus (arrowheads). 1000x.

TABLE 5: Frequency of DNA damage (%) in isolated cells of different tissues of bighead carp exposed to different levels of acetochlor.

Parameters/days	Groups/treatments			
	T0 (0.0 µg/L)	T1 (300 µg/L)	T2 (400 µg/L)	T3 (500 µg/L)
Hepatocyte (%)				
12	2.35 ± 0.04	2.37 ± 0.05	2.39 ± 0.03	4.42 ± 0.34*
24	2.37 ± 0.09	2.42 ± 0.06	3.47 ± 0.16*	4.52 ± 0.29*
36	2.43 ± 0.02	2.56 ± 0.02	3.70 ± 0.27*	5.84 ± 0.22*
Kidney cells (%)				
12	2.15 ± 0.14	2.25 ± 0.11	2.31 ± 0.14	3.21 ± 0.19*
24	2.17 ± 0.13	2.23 ± 0.16	3.75 ± 0.19*	3.59 ± 0.28*
36	2.22 ± 0.11	2.56 ± 0.18	3.87 ± 0.25*	4.81 ± 0.31*
Gills cells (%)				
12	1.15 ± 0.09	1.18 ± 0.04	1.39 ± 0.05	2.03 ± 0.14*
24	1.17 ± 0.07	1.23 ± 0.05	2.32 ± 0.02*	2.45 ± 0.19*
36	1.13 ± 0.08	1.16 ± 0.22	2.70 ± 0.04*	2.89 ± 0.12*
Brain cells (%)				
12	1.31 ± 0.11	1.33 ± 0.02	1.39 ± 0.11	1.41 ± 0.01
24	1.27 ± 0.12	1.31 ± 0.08	1.41 ± 0.03	2.33 ± 0.18*
36	1.29 ± 0.09	1.37 ± 0.008	3.11 ± 0.16*	2.67 ± 0.13*

Mean ± SE values with asterisks in a row vary significantly ($P \leq 0.05$) as compared to the control group.

ROS are naturally formed in an aerobic environment and, at physiological levels, are suggestive of oxidative eustress, or low-level oxidative stress [54]; however, ROS formed under pathological conditions and resulting in increased ROS indicates oxidative distress [55]. Enzymatic and nonenzymatic antioxidants are important for maintaining the oxidative eustress balance and redox status and provide a defense against ROS formation [54, 56]. Pesticides/herbicides are known to induce ROS and lead to oxidative stress in fish [57–60]. The antioxidant enzymes (CAT, SOD, GST, and GPx) inhibit oxidative stress, and the actions of these enzymes are usually used to monitor the risk of pesticides/herbicides [61]. Glutathione reductase is also a suitable biomarker for assessing the effect of pesticides/herbicides on aquatic organisms [60, 62].

Increased oxidative stress found in the present study could have resulted from the negative impact of contaminants linked with the generation of oxidative stress. An

increase in ROS may result from stress caused by the contaminant on intracellular constituents’ modification, defense system activity, and ROS-based signaling [3, 63]. ROS production is a natural cellular activity that is involved in varied aspects of cellular signaling, as well as in the defense mechanism of the immune system. In the present study, excessive ROS production could have resulted from acetochlor treatment that could have rendered severe damage to cellular macromolecules, such as proteins, lipids, and DNA, resulting in detrimental effects on cells [24, 33, 41, 64]. In the cells, altered nuclear processes reduce metabolic activity, induce cell membrane leakage or blockage, and decrease cell proliferation and viability; thus, oxidative stress is an important factor contributing to cell and tissue damage [49, 52].

Significantly ($P \leq 0.05$) increased oxidative stress parameters were observed in the liver and the brain of fish treated with acetochlor, in the present study. Increased oxidative stress in the brain may be the effect of the induction of

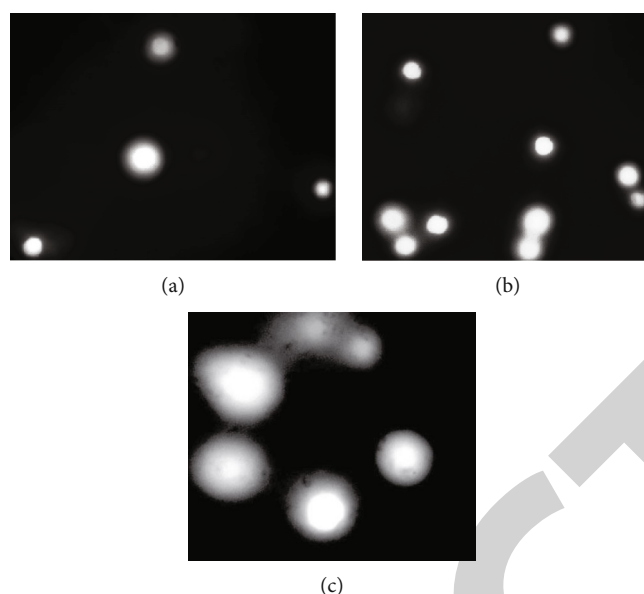


FIGURE 3: Comet assay showing DNA damage in isolated cells of the liver, kidneys, gills, and brain of fish treated with acetochlor at 300 $\mu\text{g/L}$ (a), 400 $\mu\text{g/L}$ (b), and 400 $\mu\text{g/L}$ (c). Note the frequency and intensity is increasing with the dose of acetochlor is increasing.

neurotransmitters and stress hormones released in the brain due to acetochlor treatment [65, 66]. In the current study, total protein and GSH contents in various tissues of bighead carp could be attributed to reduced tissue activities, higher usage of energy/body proteins to counteract oxidative stress, and lower protein levels in the body and tissues. It has been reported that various poisonous substances are liable for the diminution of body proteins, as well as distinct tissues of fish (*Oreochromis spilurus*, *Mystus vittatus*, *Channa punctatus*, and *Labeo rohita*), such as the liver, brain, kidneys, and gills [3, 9, 63].

Other possibility of raised oxidative stress in the present study could be the formation of free radicals as a result of toxicity induced by the acetochlor [67, 68]. The toxicant-mediated pathways can greatly increase ROS formation and excessive levels of free radicals, which might affect the metabolism [69–71]. The oxidative stress in the liver that might be the source of free radical's formation, those could interfere with intracellular signal transmission and regulation of genes, resulting in inflammation in the liver [72]. Disturbance in tissue proteins/antioxidant enzymes and production of ROS could lead to liver inflammation [70, 73, 74]. Various studies have reported different synthetic chemicals/pesticides that cause increased release of free radicals like ROS, resulting in increased induction of oxidative stress in various tissues [75], ultimately leading to activation and depletion of the body's defense responses (GSH, glutathione-S-transferase, SOD, and CAT) in exposed organisms [24, 63, 65]. Previously, it was recorded that acetochlor reduced the concentrations of different tissue proteins/antioxidant parameters like SOD and GSH at very low doses.

Moreover, acetochlor altered and damaged HepG2 cells via triggering apoptosis signals and increased deposition of calcium in cells, reduced mitochondrial transmembrane potential, and a lower quantity of ATP [24]. ROS can cause damage to biomolecules leading to cell and tissue injury

[76]. Moreover, it is recorded that acetochlor/herbicides mainly induce an increased lipid peroxidation and ROS production process, leading to damage to DNA, proteins, lipids, and carbohydrates that consequently affect the immune functions [77–82].

The nuclear and morphological alterations in RBCs of bighead carp like condensed nuclei, micronuclei, pear shape erythrocyte, and spherocyte could be due to the toxic effects of herbicides on hematopoietic tissues. Previous studies have recorded the occurrence of nuclear and morphological alterations in erythrocytes of humans [83, 84], birds [7, 9], and fish [21, 24, 25, 63, 85] might be related to mitochondrial damage.

5. Conclusions

This study concluded that acetochlor leads to significantly ($P < 0.05$) higher morphological and nuclear abnormalities in erythrocytes in treated fish. There was significantly increased DNA damage in isolated cells of treated fish's gills, liver, brain, and kidneys. The results on oxidative stress showed a higher quantity of different oxidative biomarkers and a significantly ($P < 0.05$) lower concentration of various cellular proteins/antioxidant enzymes in the liver, gills, kidneys, and brain of treated bighead carp compared to unexposed fish. Our research findings concluded that acetochlor causes harmful patho-biochemical changes in different tissues of the bighead carp.

6. Limitations of the Study

As this is a laboratory study and carried out within limited sources, this type of study is carried out under field conditions involving various other carp species to carry out a broad conclusion. Moreover, remedial measures are searched out for the amelioration of acetochlor and other insecticides/pesticides/herbicides so that farmers could get more profit from fish farming.

Abbreviations

ANOVA: Analysis of variance
CAT: Catalase
GSH: Reduced glutathione
POD: Peroxidase
ROS: Reactive oxygen species
SE: Standard error
SOD: Superoxide dismutase
TBARS: Thiobarbituric acid reactive substance.

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

Authors' Contributions

Abdul Ghaffar and Riaz Hussain planned and designed the research work. Riaz Hussain and Yasir Mahmood executed the study and obtained the data. Riaz Hussain and Ahrar Khan analyzed the collected data. Riaz Hussain, Ahrar Khan, and Khalid Mehmood interpreted the data. Riaz Hussain, Ahrar Khan, Sadia Nawaz, and Yasir Mahmood prepared the manuscript paper. All authors read and approved the final version of the manuscript.

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